

School of Computer Science Engineering and Technology

Course-B. Tech	Type- Core
Course Code- CSET207	Course Name- COMPUTER NETWORKS
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CO-Mapping

	CO1	CO2	CO3
Q1	✓		
Q2	✓		
Q3	✓		
Q4	✓	✓	

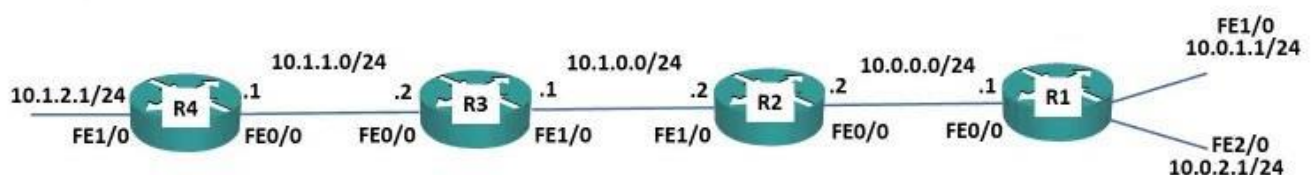
Objective – Interior Gateway Routing and Static Routes: Understanding Default Routing, Static Routing.

When a router receives an IP packet through any of its interfaces, the router examines the packet's destination IP address, the optimal path to reach this destination is added to the routing table. Metrics are used to determine the optimal path to reach a destination IP address through static and dynamic routing protocols. These metrics are standard measurements or vectors that give a quantitative value measure for the distance to a given network. A common metric can either be hop count, bandwidth, delay, current load on the path, reliability, or the combination of two or more of these standards.

Routers forward packets using either route information from route table entries that you manually configure or the route information that is calculated using dynamic routing algorithms. Static routes, which define explicit paths between two routers, cannot be automatically updated; you must manually reconfigure static routes when network changes occur. Static routes use less bandwidth than dynamic routes. No CPU cycles are used to calculate and analyze routing updates.

• Routes on R1:

```
ip route 10.1.0.0 255.255.255.0 10.0.0.2
ip route 10.1.1.0 255.255.255.0 10.0.0.2
ip route 10.1.2.0 255.255.255.0 10.0.0.2
```



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You can supplement dynamic routes with static routes where appropriate. You can redistribute static routes into dynamic routing algorithms, but you cannot redistribute routing information calculated by dynamic routing algorithms into the static routing table.

You should use static routes in environments where network traffic is predictable and where the network design is simple. You should not use static routes in large, constantly changing networks because static routes cannot react to network changes. Most networks use dynamic routes to communicate between routers but may have one or two static routes configured for special cases. Static routes are also useful for specifying a gateway of last resort (a default router to which all unroutable packets are sent).

```
Router_B#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/24 is subnetted, 1 subnets
S       10.0.0.0 [1/0] via 192.168.1.1
  20.0.0.0/24 is subnetted, 1 subnets
C       20.0.0.0 is directly connected, FastEthernet0/0
C       192.168.1.0/24 is directly connected, Serial2/0
Router_B#
```

Administrative Distance: An administrative distance is the metric used by routers to choose the best path when there are two or more routes to the same destination from two different routing protocols. An administrative distance guides the selection of one routing protocol (or static route) over another, when more than one protocol adds the same route to the unicast routing table. Each routing protocol is prioritized in order of most to least reliable using an administrative distance value.

Static routes have a default administrative distance of 1. A router prefers a static route to a dynamic route because the router considers a route with a low number to be the shortest. If you want a dynamic route to override a static route, you can specify an administrative distance for the static route. For example, if you have two dynamic routes with an administrative distance of 120, you will specify an administrative distance that is greater than 120 for the static route if you want the dynamic route to override the static route.

Directly Connected Static Routes: You need to specify only the output interface (the interface on which all packets are sent to the destination network) in a directly connected static route. The router assumes the destination is directly attached to the output interface and the packet destination is used as the next hop address. The next hop can be an interface, only for point-to-point interfaces. For broadcast interfaces, the next hop must be an IPv4 address.

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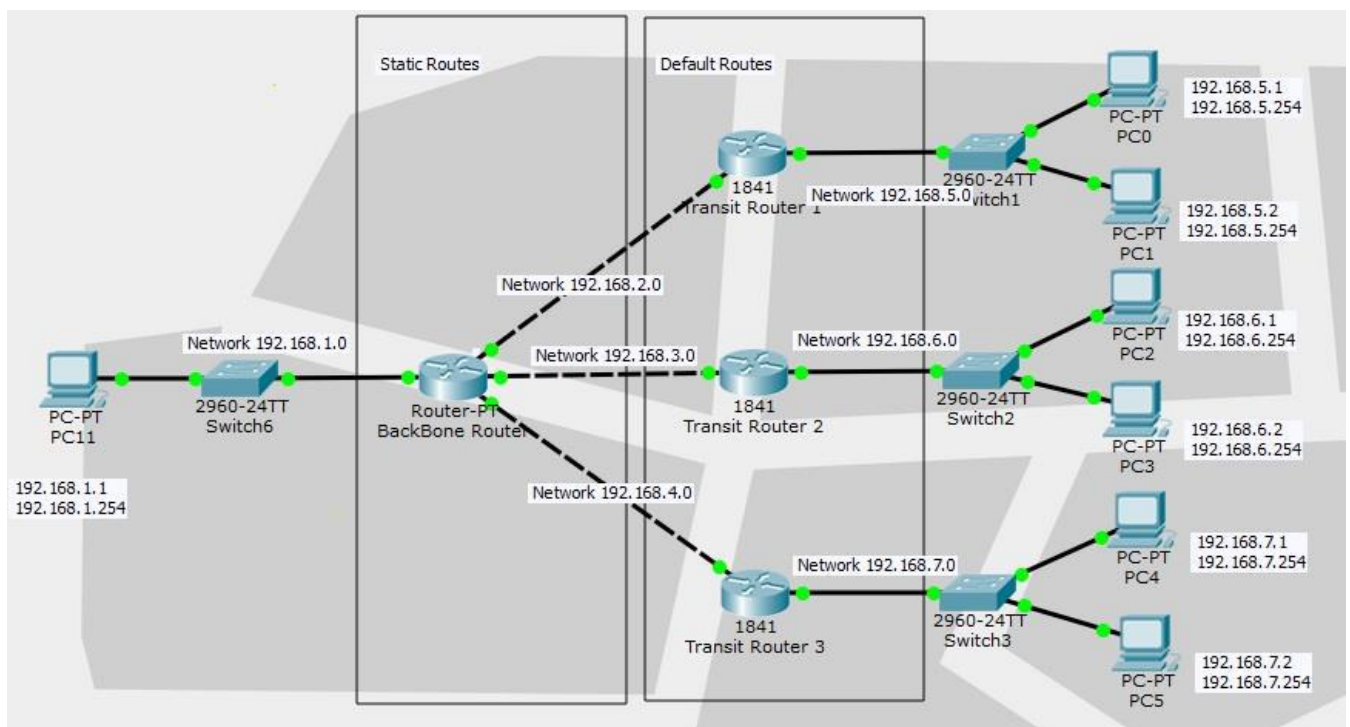
Fully Specified Static Routes: You must specify either the output interface (the interface on which all packets are sent to the destination network) or the next hop address in a fully specified static route. You can use a fully specified static route when the output interface is a multi-access interface, and you need to identify the next-hop address. The next-hop address must be directly attached to the specified output interface.

Floating Static Routes: A floating static route is a static route that the router uses to back up a dynamic route. You must configure a floating static route with a higher administrative distance than the dynamic route that it backs up. In this instance, the router prefers a dynamic route to a floating static route. You can use a floating static route as a replacement if the dynamic route is lost.

Prerequisites for Static Routing: If the next-hop address for a static route is unreachable, the static route will not be added to the unicast routing table.

Guidelines and Limitations: You can specify an interface as the next-hop address for a static route only for point-to-point interfaces such as GRE tunnels.

Task 1: Setup a network topology in Packet Tracer, using one backbone and three transit routers. The network classification and addressing scheme is marked below:



Task 2: Configure Backbone and Transit Routers for addressing.

BackBone Router

```
Router>enable
Router#config t
Router(config)#interface fastethernet8/0
Router(config-if)#ip address 192.168.1.254 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface fastethernet0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface fastethernet1/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
```

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```
Router(config-if)#interface fastethernet7/0
Router(config-if)#ip address 192.168.4.1 255.255.255.0
Router(config-if)#no shutdown
```

Transit Router 1

```
Router>enable
Router#config t
Router(config)#interface fastethernet0/0
Router(config-if)#ip address 192.168.2.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface fastethernet0/1
Router(config-if)#ip address 192.168.5.254 255.255.255.0
Router(config-if)#no shutdown
```

Transit Router 2

```
Router>enable
Router#config t
Router(config)#interface fastethernet0/0
Router(config-if)#ip address 192.168.3.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface fastethernet0/1
Router(config-if)#ip address 192.168.6.254 255.255.255.0 Router(config-if)#no shutdown
```

Transit Router 3

```
Router>enable
Router#config t
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.4.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface fastethernet 0/1
Router(config-if)#ip address 192.168.7.254 255.255.255.0
Router(config-if)#no shutdown
```

Task 3: Configure Static Routes on the BackBone Router.

```
Router>enable
Router#config t
Router(config)#ip route 192.168.5.0 255.255.255.0 192.168.2.2
Router(config)#ip route 192.168.6.0 255.255.255.0 192.168.3.2
Router(config)#ip route 192.168.7.0 255.255.255.0 192.168.4.2
```

Task 4: Configure Default Routes on Transit Routers.

Transit Router 1

```
Router>enable
Router#config t
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.2.1
```

Transit Router 2

```
Router>enable
Router#config t
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.3.1
```

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Transit Router 3

Router>enable

Router#config t

Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.4.1

Task 6: Verify Configuration.

Router#show ip route

Router#show ip route 192.168.1.0